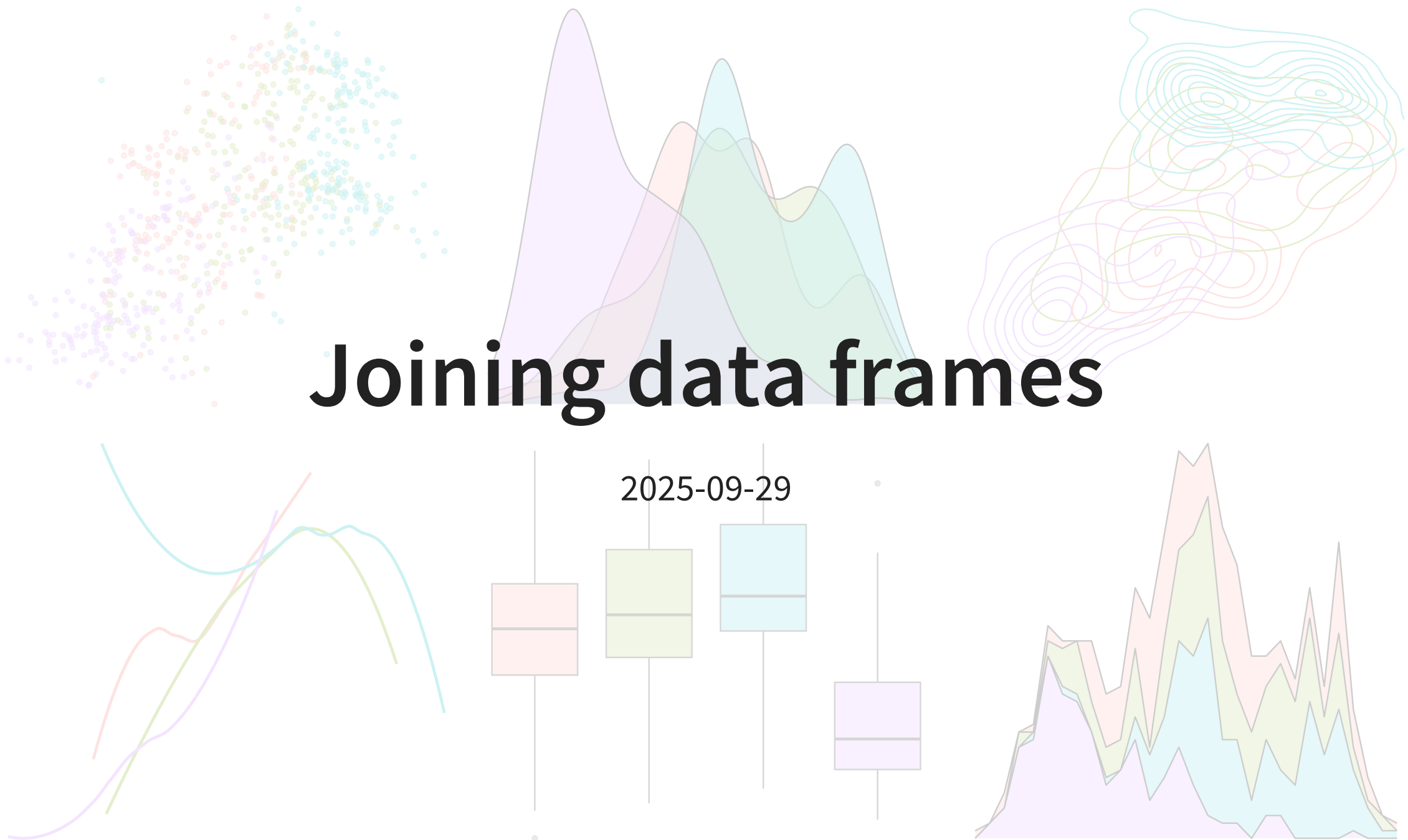


Joining data frames



Housekeeping

- Problem Set 2 graded! Please look at feedback on Gradescope.
- Problem Set 3 due tonight!

Joining data frames

Assume we have two data frames, `df1` and `df2`. There are some shared variable(s) across the two.

We want to combine `df1` and `df2` into one single data frame. The `dplyr` code will look something like:

```
1 xxx_join(df1, df2)
```

- How do we combine `df1` and `df2` them? The following mutating joins give us rules:
 - `left_join()`: keep all cases from `df1`
 - `right_join()`: keep all cases from `df2`
 - `full_join()`: keep all cases from both `df1` and `df2`
 - `inner_join()`: keep all cases from `df1` where there are matching values in `df2` (more complicated in the scenario of multiple matches)

I know this is confusing! Examples will help clarify!

Setup

For the next few slides...

1 df1

ID	name
1	x1
2	x2
3	x3

1 df2

ID	grade
1	y1
2	y2
4	y4

I recommend writing these two down, as we will work with them in the next few slides.

Note! The variable **ID is shared between these two data frames. We will “join on” **ID**.**

left_join(df1, df2)

Adds columns to **df1** from **df2**, matching the cases in **df1**. Retains ALL cases from left data frame **df1**, and adds **NA** for variables from **df2** that have no match.

```
1 df1
```

ID	name
1	x1
2	x2
3	x3

```
1 df2
```

ID	grade
1	y1
2	y2
4	y4

```
1 left_join(df1, df2, by = "ID")
```

ID	name	grade
1	x1	y1
2	x2	y2
3	x3	NA

left_join(x, y)

1	x1	1	y1
2	x2	2	y2
3	x3	4	y4

Think of left data frame as fixed.

right_join(df1, df2)

Adds columns to `df1` from `df2`, matching the cases in `df2`. Retains ALL cases from right data frame `df2`, and adds `NA` for variables from `df1` that have no match.

```
1 df1
```

ID	name
1	x1
2	x2
3	x3

```
1 df2
```

ID	grade
1	y1
2	y2
4	y4

```
1 right_join(df1, df2, by = "ID")
```

ID	name	grade
1	x1	y1
2	x2	y2
4	NA	y4

right_join(x, y)

1	x1
2	x2
3	x3

1	y1
2	y2
4	y4

Think of right data frame as fixed

full_join(df1, df2)

Adds columns to **df1** from **df2**, *fully* combining the two data frames, matching when possible. Retains ALL cases from **df1** AND **df2**, and adds **NAs** as necessary.

```
1 df1
```

ID	name
1	x1
2	x2
3	x3

```
1 df2
```

ID	grade
1	y1
2	y2
4	y4

```
1 full_join(df1, df2, by = "ID")
```

ID	name	grade
1	x1	y1
2	x2	y2
3	x3	NA
4	NA	y4

full_join(x, y)



No chance for losing information.

inner_join(df1, df2)

Adds columns to `df1` from `df2`, retaining only cases that exist in both `df1` AND `df2`.

```
1 df1
```

ID	name
1	x1
2	x2
3	x3

```
1 df2
```

ID	grade
1	y1
2	y2
4	y4

```
1 inner_join(df1, df2, by = "ID")
```

ID	name	grade
1	x1	y1
2	x2	y2

inner_join(x, y)

1	x1	1	y1
2	x2	2	y2
3	x3	4	y4

Comprehension check

1. In which of the joins are **NAs** possibly introduced? In which of the joins are **NAs** never going to be introduced?
2. If we have some **df1** and **df2**, in what scenario will the result of the four joins be the same? (i.e. what has to be true about **df1** and **df2**?)
3. What do you think we call these “mutating” joins? *Think about where we’ve seen that verb before!*

- Note: it’s possible to join on more than one variable!

Example

We have data on fishery harvests (in tons) by countries from 2016, stored in data frame `fish`:

What does this code do?

```
1 fish |>
2   slice(1:8)
```

country	capture	aquaculture
Afghanistan	1000	1200
Albania	7886	950
Algeria	95000	1361
American Samoa	3047	20
Andorra	0	0
Angola	486490	655
Antigua and Barbuda	3000	10
Argentina	755226	3673

Suppose I would like to explore the data on a continent level. We have another data frame called `continent` that looks like this:

```
1 continents |>
2   slice(1:8)
```

Country	continent
Afghanistan	Asia
Åland Islands	Europe
Albania	Europe
Algeria	Africa
American Samoa	Oceania
Andorra	Europe
Angola	Africa
Anguilla	Americas

Discuss

`fish` data frame snapshot:

country	capture	aquaculture
Afghanistan	1000	1200
Albania	7886	950
Algeria	95000	1361
American Samoa	3047	20
Andorra	0	0

`continents` data frame snapshot:

Country	continent
Afghanistan	Asia
Åland Islands	Europe
Albania	Europe
Algeria	Africa
American Samoa	Oceania

- We want to keep all rows and columns from `fish` and add a column for corresponding continents. Which join function should we use?
- We want to keep only the rows from `fish` for which we have a corresponding continent and add a column for corresponding continents. Which join function should we use?
- Which variable(s) are we joining on?

Example (cont.)

Question 1:

```
1 left_join(fish, continents,  
2           by = c("country" = "Country")) |>  
3 slice(1:8)
```

country	capture	aquaculture	continent
Afghanistan	1000	1200	Asia
Albania	7886	950	Europe
Algeria	95000	1361	Africa
American Samoa	3047	20	Oceania
Andorra	0	0	Europe
Angola	486490	655	Africa
Antigua and Barbuda	3000	10	NA
Argentina	755226	3673	Americas

- Notice the **NA**
- Notice we don't have two copies of same **country/Country** variable!

Question 2:

```
1 inner_join(fish, continents,  
2            by = c("country" = "Country")) |>  
3 slice(1:8)
```

country	capture	aquaculture	continent
Afghanistan	1000	1200	Asia
Albania	7886	950	Europe
Algeria	95000	1361	Africa
American Samoa	3047	20	Oceania
Andorra	0	0	Europe
Angola	486490	655	Africa
Argentina	755226	3673	Americas
Armenia	3758	16381	Asia

Piping into `xxx_join()`

In the following line of code, when would an NA be introduced? (Answer forthcoming.)

```
1 right_join(fish, continents, by = c("country" = "Country"))
```

- Remember in data wrangling, we pipe from a previous line of code directly into another
- We can pipe directly into the join functions! The following achieves the same as above:

```
1 fish |>  
2   right_join(continents, by = c("country" = "Country"))
```

Answer to previous question

NAs would be introduced if there is a country in `continents` that does not appear in `fish`. See the last 50-ish cases:

Show

10

 entries

Search:

	country	capture	aquaculture	continent
1	Afghanistan	1000	1200	Asia
2	Albania	7886	950	Europe
3	Algeria	95000	1361	Africa
4	American Samoa	3047	20	Oceania
5	Andorra	0	0	Europe
6	Angola	486490	655	Africa
7	Argentina	755226	3673	Americas
8	Armenia	3758	16381	Asia